

Quiz navigation

1 2 3 4 5

Finish review

Course Home

Course Info

Syllabus

Participants list

Grade/Attendance

Offline-Attendance

Grades

Students Notifications

Others

Activities/Resources



Engineering Mathematics

12Week [20 November - 26 November]

Quiz 12

Started on	2023-11-27 06:57
State	Finished
Completed on	2023-11-27 06:57
Time taken	36 secs
Grade	4.00 out of 5.00 (80%)

Question 1

Correct

Mark 1.00 out of 1.00

Fill in the gaps in the multiplication

$$\begin{bmatrix} 1 & 3 & A \\ 0 & 5 & -4 \\ 8 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & B \\ -2 & 0 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} -11 & 3 \\ C & 0 \\ 5 & D \end{bmatrix} \text{ and among}$$

the numbers below choose the correct value of $A + B + C + D$

Select one:

- ☐ 1
- ☐ 2
- ☒ 3 ✓
- ☐ 4

Your answer is correct.

The correct answer is: 3

Question 2

Correct

Mark 1.00 out of 1.00

Below we find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix}.$$

Fill in the gaps in the solution and choose the correct value of

$$A + B + C + D.$$

$$\begin{aligned} [A|I] &= \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{(1) \times (-1) + (2)} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & A & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \\ &\xrightarrow{(3) \times (-1) + (2)} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & A & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & B \end{array} \right] \xrightarrow{(3) \times (-2) + (2)} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & C & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 & B \end{array} \right] \\ &\xrightarrow{(2) \times (-1) + (1)} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & D & -1 \\ 0 & 1 & 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & -1 & 1 & B \end{array} \right] \end{aligned}$$

Select one:

- ☒ 0 ✓
- ☐ 1
- ☐ 2
- ☐ 3

Your answer is correct.

The correct answer is: 0

Question 3

Incorrect

Mark 0.00 out of 1.00

Among the vectors below choose one which is a solution of the system

$$x'(t) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} x(t).$$

Select one:

☐ $x(t) = \begin{bmatrix} 0 \\ 1 \\ t^2 \end{bmatrix}$

☒ $x(t) = \begin{bmatrix} e^t \\ 2e^t \\ e^t \end{bmatrix}$ ✖

☐ $x(t) = \begin{bmatrix} 0 \\ e^t \\ -3e^t \end{bmatrix}$

☐ $x(t) = \begin{bmatrix} 1 \\ e^t \\ t^2 \end{bmatrix}$

Your answer is incorrect.

The correct answer is: $x(t) = \begin{bmatrix} 0 \\ e^t \\ -3e^t \end{bmatrix}$

Question 4

Correct

Mark 1.00 out of 1.00

Rewrite the equation $y''' - y' + y = \cos t$ in normal form

$x'(t) = A(t)x(t) + f(t)$ using the notations

$$x(t) = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \text{where} \quad x_1 = y, \quad x_2 = y', \quad x_3 = y''.$$

Choose $A(t)$ and $f(t)$ among the answers below.

Select one or more:

☐ $A(t) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

☒ $A(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$ ✓

☐ $A(t) = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix}$

☒ $f(t) = \begin{bmatrix} 0 \\ 0 \\ \cos t \end{bmatrix}$ ✓

☐ $f(t) = \begin{bmatrix} \cos t \\ 0 \\ 0 \end{bmatrix}$

☐ $f(t) = \begin{bmatrix} 1 \\ \cos t \\ 0 \end{bmatrix}$

Your answer is correct.

The correct answers are:

$$A(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix},$$

$$f(t) = \begin{bmatrix} 0 \\ 0 \\ \cos t \end{bmatrix}$$

Question 5

Correct

Mark 1.00 out of 1.00

For which vector f below the vector

$$x_p = \begin{bmatrix} 5t + 1 \\ 2t \\ 4t + 2 \end{bmatrix}$$

is a

particular solution to the system

$$x' = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} x + f?$$

Select one:

☐ $f = \begin{bmatrix} 2t \\ -1 \\ -3t \end{bmatrix}$

☐ $f = \begin{bmatrix} 3t \\ 0 \\ 2t \end{bmatrix}$

☒ $f = \begin{bmatrix} -9t \\ 0 \\ -18t \end{bmatrix}$ ✓

☐ $f = \begin{bmatrix} 12t \\ 1 \\ -4t \end{bmatrix}$

Your answer is correct.

The correct answer is:

$$f = \begin{bmatrix} -9t \\ 0 \\ -18t \end{bmatrix}$$